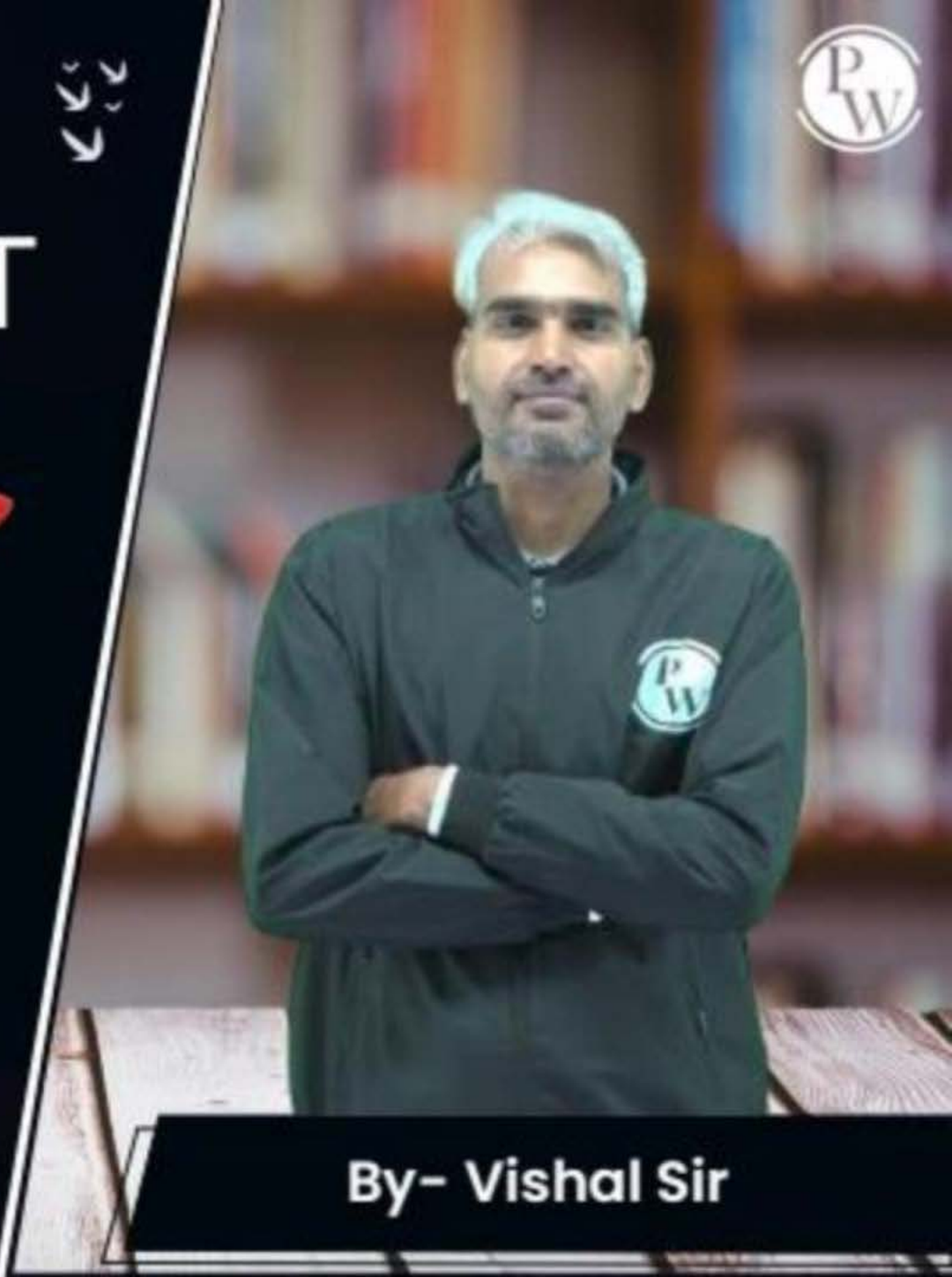
Computer Science & IT



Database Management System

Query Languages

Lecture No. 07



By- Vishal Sir

Recap of Previous Lecture



Topic Nested queries

Topic IN, ANY, ALL and EXISTS operators

Topic Execution of Nested queries

Slide

Topics to be Covered



- Topic: Execution of Nested queries
- Topic: AS clause
- Topic: WITH clause
- Topic: Comparison with NULL
- Topic: Comparison with regular expression

Slide



Topic : Nested queries

Nested Query (Query within query)

Independent Nested query

When inner query does not depend upon Outer query
{i.e. it can be executed independently}

Correlated Nested query

Inner query uses the attribute from the relation specified in the outer query



Topic : Independent nested query

eg: Select Sid
From Student
Where (marks = (Select max(marks)
From Student))

Outer query

inner query

it can be executed independently
i.e. Independent query

Note: If inner query is independent query then, inner query will be executed first.



Topic : Correlated nested query

Select *
From R
Where some operator (Select *
From S
Where (R.A = S.B))

outer query

inner query

because of this it is a Correlated query



Topic : Correlated nested query

Order of Execution

6 — Select *

1 — From R

5 ← Where Some operator

{ (Select * — 4

From S — 2

Where (R.A = S.B) — 3)

for each tuple of Outer relation(R) this three steps will be repeated w.r.t. Complete inner relation(S)

NOTE :- ① Best used operators w.r.t. independent nested query are IN, ANY, ALL

② Best used operator w.r.t. Correlated nested query is "EXISTS"



Topic : IN operator

- IN operator is used to check whether the tuple under consideration is present in a given set of tuples or not.

tuple under consideration

Set of tuples

e.g. X IN {2, 4, 6, 8, 9}

if X = 4, then IN will return true

if X = 5, then IN will return false

Complement of IN is "NOT IN"

Eg:

Supplier (Sid, Sname, Rating)

Parts (Pid, Pname, Color)

Catalog (Sid, Pid, Cost)

Catalog

S_1	P_1
S_1	P_2
S_2	P_2
S_3	P_1

Parts

P_1	Red
P_2	Green
P_3	Red

Retrieve Sids of the suppliers who supplied some Red Color Parts

S_1, S_3

Soln:

```
Select Sid
From Catalog
Where Pid IN { Select Pid
               From Parts
               Where (Color = 'RED') }
```

{ P_1, P_3 }

Output of inner query will provide us the set of tuples

→ If inner query result is empty (i.e. set of tuples is empty) then IN operator will always return false

"NOT IN" is Complement of IN

∴ "NOT IN" will return true if inner query result is EMPTY



Topic : ANY operator

ANY operator is used along with other Comparision Operators
{i.e. =, <, >, ≤, ≥, <> }

Not equal

"ANY" operator will return true if at least one tuple in set of tuples satisfy the Comparison Condition.

set of tuples

Eg: $X \leq ANY \{2, 4, 6, 8, 9\}$

if X=5, then ANY operator will return true

if X=15, then ANY operator will return false

Note: "=ANY" is equivalent to "IN" operator

Note:- If inner query result is empty then there is no tuple in the set of tuples to satisfy the Comparision Condition

∴ If inner query result is empty then ANY operator will always return false



Topic : ALL operator

- Just like "ANY" operator, "ALL" operator is also used along with other Comparision Condition

- ALL operator returns False only if at least one tuple in the set of tuples fails the comparison condition
  - ∴ If there is no tuple in the set of tuples then "ALL" operator will always return true

eg: $X > ALL \{2, 4, 6, 8, 9\}$ set of tuples

if $X = 20$, then ALL returns true

if $X = 7$, then ALL will return false

Note: "NOT IN" is equivalent to "<> ALL"



Topic : EXISTS operator

- If inner query result is non-empty then "EXISTS" operator will return true

- If inner query result is Empty then "EXISTS" operator will return false

- Complement of "EXISTS" is "NOT EXISTS"

HW:

#e.g.

```
SELECT C.sid
FROM Catalog C
WHERE EXISTS ( SELECT *
               FROM Parts P
               WHERE (P.Pid=C.Pid AND P.color='RED'))
```

Catalog

Sid	Pid
S_1	P_1
S_1	P_2
S_2	P_2
S_3	P_3

Parts

Pid	Color
P_1	Red
P_2	Green
P_3	Red

What will be the output of above query if it is executed W.r.t given instances of Catalog & Parts.

O/P =

Sid
S_1
S_3

#e.g. SELECT C.sid

FROM Catalog C

WHERE EXISTS (SELECT * FROM Parts P WHERE (P.Pid=C.Pid AND P.color='RED'))

S_1

Catalog table is renamed as C

C

S_1	P_1

Parts P

Pid	Color
P_1	Red
P_2	Green
P_3	Red

P_1 Red

~~P_2 Green~~

~~P_3 Red~~

Catalog

Sid	Pid
S_1	P_1
S_1	P_2
S_2	P_2
S_3	P_3

Not empty

∴ EXISTS return true for 1st tuple of Catalog

Hence Sid from the first tuple is selected

#e.g. SELECT C.sid

FROM Catalog C

WHERE EXISTS (SELECT *

FROM Parts P

WHERE (P.Pid=C.Pid AND P.color='RED'))

C

S_1	P_2

Parts

Pid	Color
P_1	Red
P_2	Green
P_3	Red

P_1 Red

P_2 Green

P_3 Red

Catalog

Sid	Pid
S_1	P_1
S_1	P_2
S_2	P_2
S_3	P_3

for 2nd tuple of Catalog table inner query result is empty.

Exists will return false

Hence Sid of 2nd tuple of Catalog is not selected

S_2

#e.g. SELECT C.sid

FROM Catalog C → S_2 P_2

WHERE EXISTS (SELECT * → P_1 Red, P_2 Green, P_3 Red

FROM Parts P

WHERE (P.Pid=C.Pid AND P.color='RED'))

false

Empty

∴ S_2 is not selected

Parts

Pid	Color
P_1	Red
P_2	Green
P_3	Red

Catalog

Sid	Pid	
S_1	P_1	✓
S_1	P_2	✓
S_2	P_2	✓
S_3	P_3	

S_3

#e.g. SELECT C.sid

FROM Catalog C

WHERE EXISTS (SELECT *

FROM Parts P

WHERE (P.Pid=C.Pid AND P.color='RED'))

Catalog

Sid	Pid
S_1	P_1
S_1	P_2
S_2	P_2
S_3	P_3

S_3	P_3
~~P_1~~	~~Red~~
~~P_2~~	~~Green~~
P_3	Red

Parts

Pid	Color
P_1	Red
P_2	Green
P_3	Red

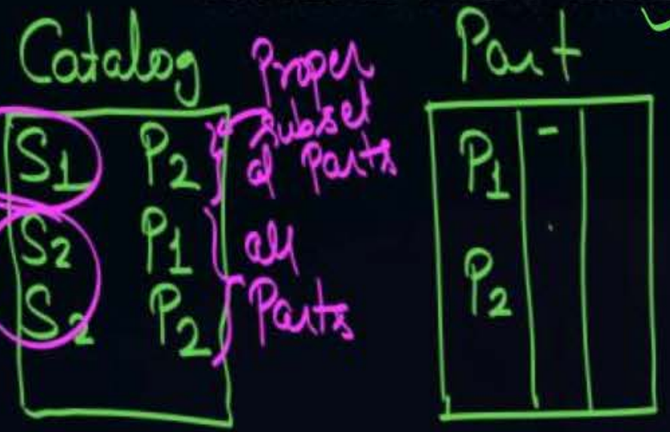
Not empty

∴ EXISTS return True

Hence Sid of 4th tuple will be selected

#e.g.

```
SELECT C1.sid
FROM Catalog C1
WHERE NOT EXISTS ( SELECT P.Pid
                   FROM Parts P
                   WHERE NOT EXISTS (SELECT C2.Sid
                                     FROM Catalog C2
                                     WHERE(C2.Pid=P.Pid AND  C2.Sid=C1.Sid)))
```



What output is produced by above SQL query:

A. Sids of suppliers who supplied some parts

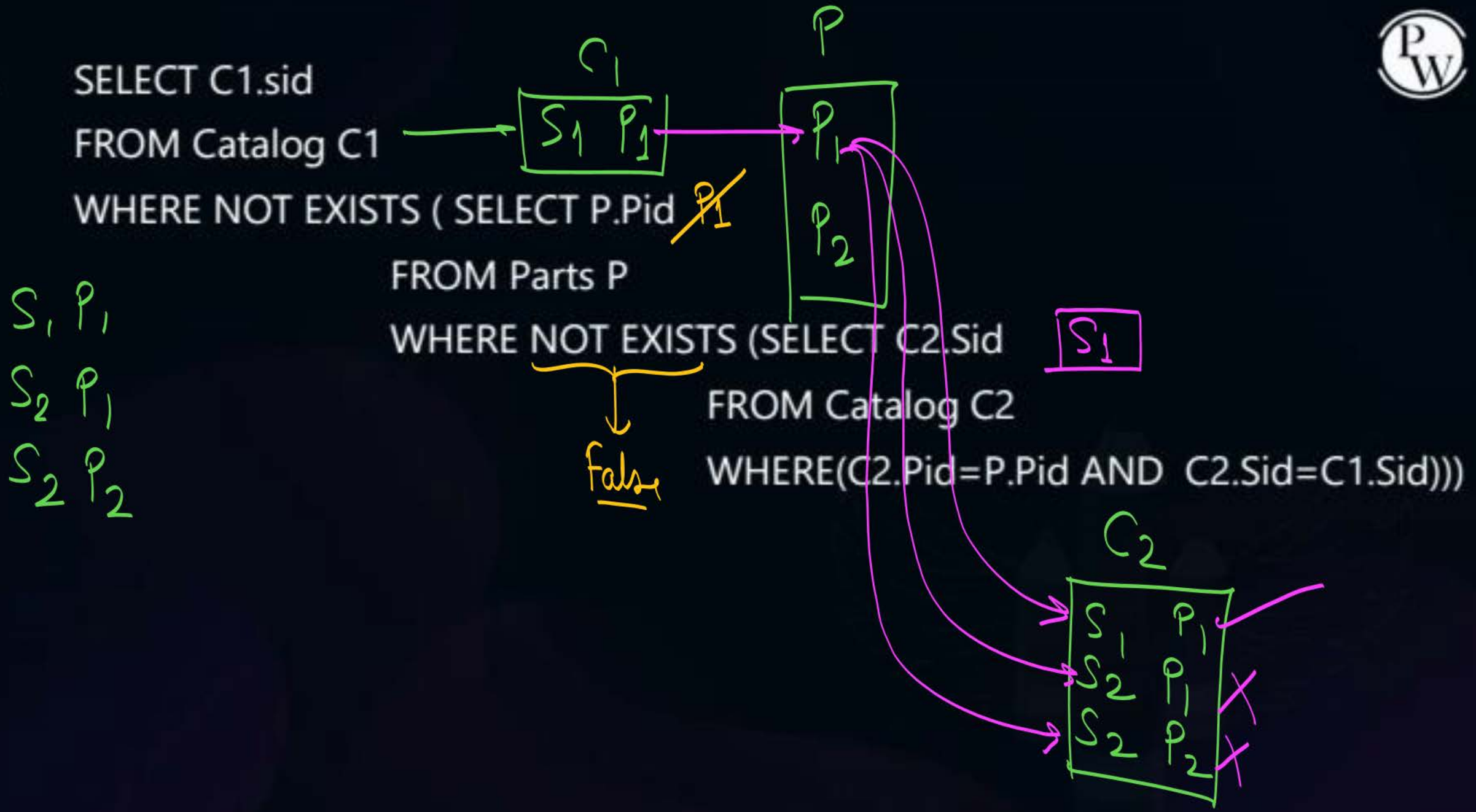
B. Sids of suppliers who supplied only proper subset of parts from all parts

C. Sids of suppliers who supplied all parts

D. Sids of suppliers who did not supply any part

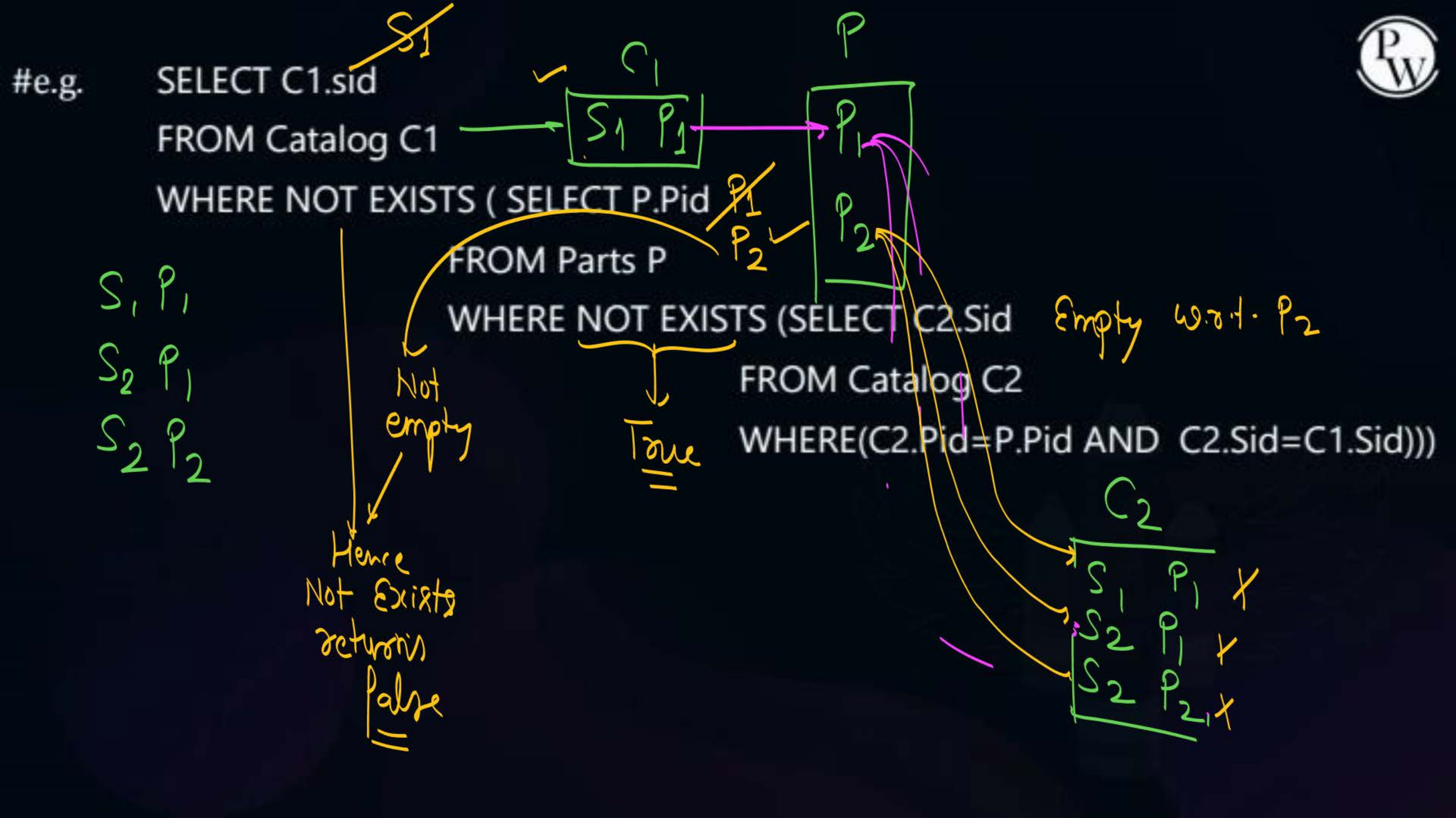
#e.g.

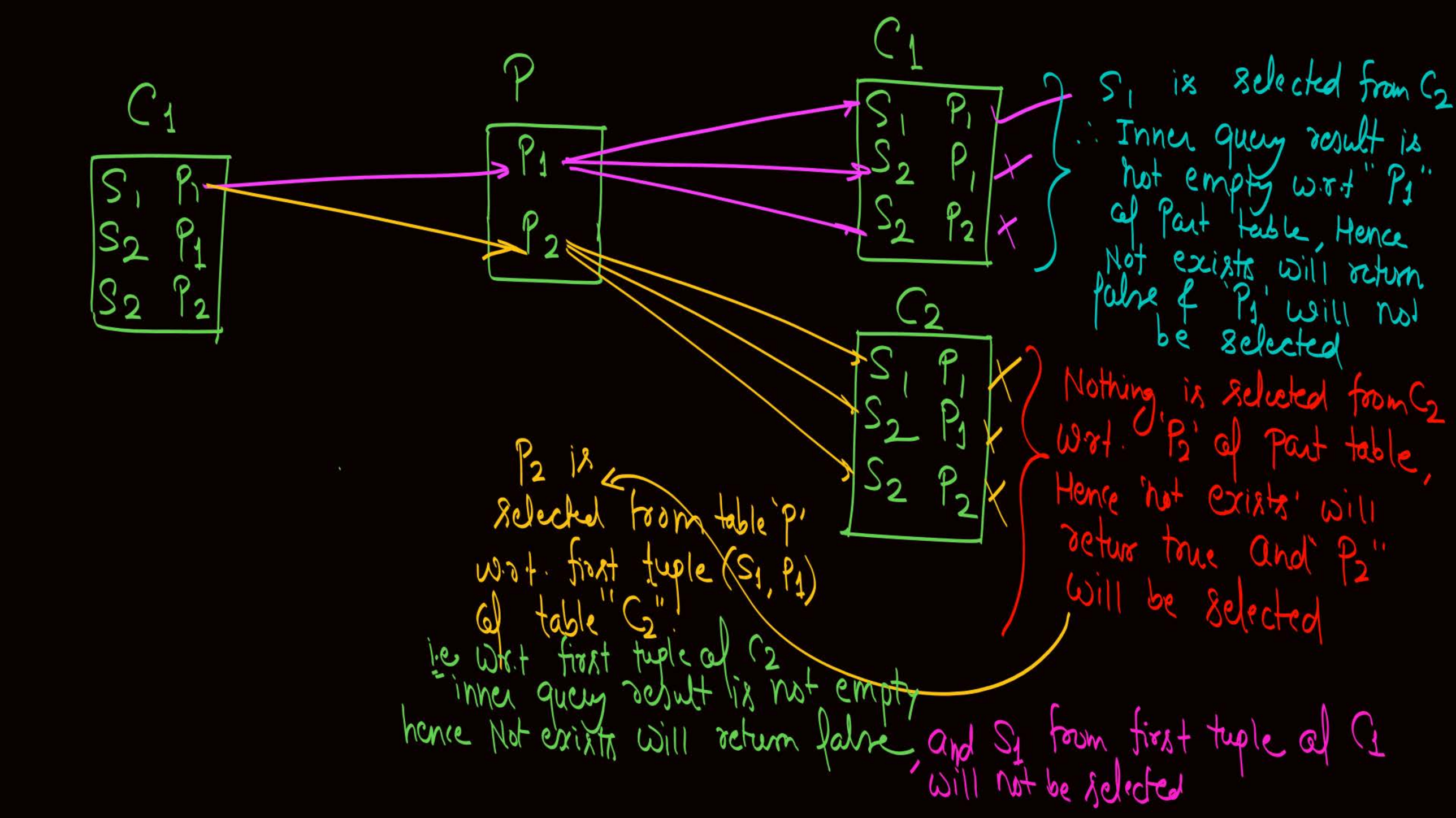
```
SELECT C1.sid
FROM Catalog C1
WHERE NOT EXISTS ( SELECT P.Pid
                   FROM Parts P
                   WHERE NOT EXISTS (SELECT C2.Sid
                                     FROM Catalog C2
                                     WHERE(C2.Pid=P.Pid AND  C2.Sid=C1.Sid)))
```

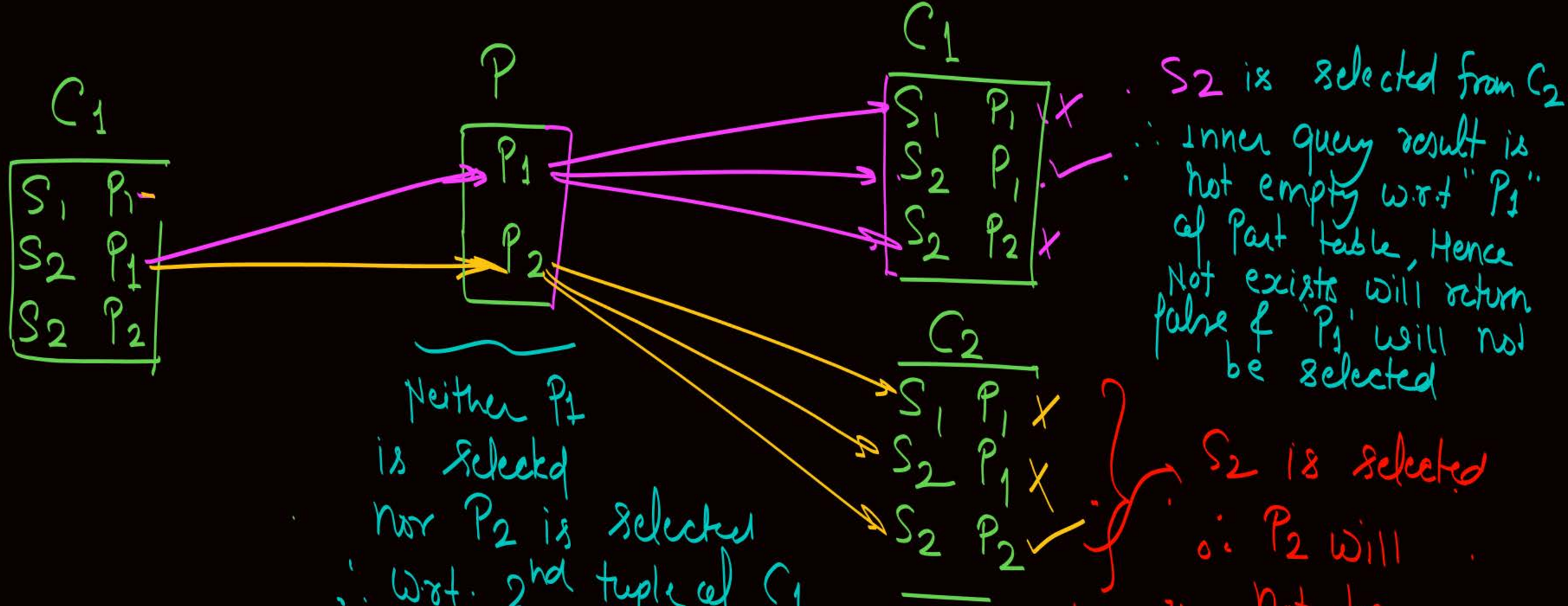


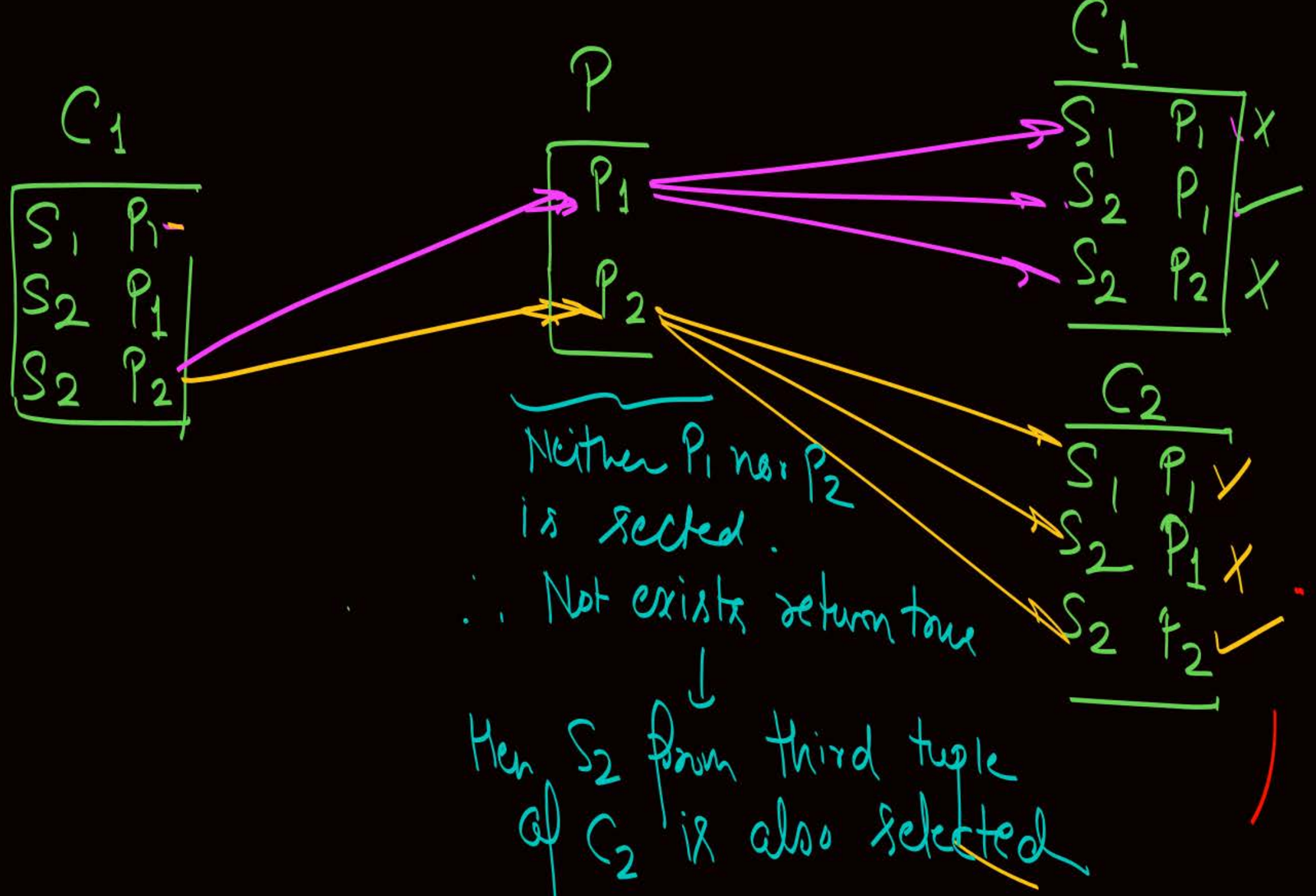
#e.g.

```
SELECT C1.sid
FROM Catalog C1
WHERE NOT EXISTS ( SELECT P.Pid
                   FROM Parts P
                   WHERE NOT EXISTS (SELECT C2.Sid
                                     FROM Catalog C2
                                     WHERE(C2.Pid=P.Pid AND C2.Sid=C1.Sid)))
```




C1
S1 P1
S2 P1
S2 P2
P
P1
P2
C1
S1 P1
S2 P1
S2 P2
S1 is selected from C2
∴ Inner query result is not empty w.r.t "P1" of Part table, Hence Not exists will return false & 'P1' will not be selected
C2
S1 P1
S2 P1
S2 P2
Nothing is selected from C2 w.r.t 'P2' of Part table, Hence 'not exists' will return true and 'P2' will be selected
P2 is selected from table 'P' w.r.t. first tuple (S1, P1) of table "C2"
i.e w.r.t first tuple of C2 inner query result is not empty hence Not exists will return false, and S1 from first tuple of C1 will not be selected


C1
S1 P1
S2 P1
S2 P2
P
P1
P2
C1
S1 P1
S2 P1
S2 P2
C2
S1 P1
S2 P1
S2 P2
∴ S2 is selected from C2
∴ inner query result is not empty w.r.t "P1" of Part table, Hence Not exists will return false & P1 will not be selected
Neither P1 is selected nor P2 is selected
∴ w.r.t. 2nd tuple of C1 inner query result is empty
Hence "Not exists" will return true
and "S2" from C1 will be selected
∴ S2 is selected
∴ P2 will not be selected


C1
S1 P1
S2 P1
S2 P2
P
P1
P2
C1
S1 P1 X
S2 P1
S2 P2 X
C2
S1 P1 ✓
S2 P1 X
S2 P2 ✓
Neither P1 nor P2 is selected.
∴ Not exists return true
Hence S2 from third tuple of C2 is also selected
∴ S1 is selected from C2
∴ Inner query result is not empty w.r.t "P1" of Part table, Hence Not exists will return false & P1 will not be selected
S2 is selected
∴ P2 will not be selected

#e.g.

```
SELECT C1.sid
FROM Catalog C1
WHERE NOT EXISTS ( SELECT P.Pid
                   FROM Parts P
                   WHERE NOT EXISTS (SELECT C2.Sid
                                     FROM Catalog C2
                                     WHERE(C2.Pid=P.Pid AND  C2.Sid=C1.Sid)))
```

on this instances of Catalog and parts if given query is executed then o/p will be

C1.Sid
S_2
S_2



What output is produced by above SQL query:

A. Sids of suppliers who supplied some parts

B. Sids of suppliers who supplied only proper subset of parts from all parts

C. Sids of suppliers who supplied all parts

D. Sids of suppliers who did not supply any part



Topic : AS clause

AS clause is used to rename a column or table with an alias.

An alias only exists for the duration of the query.

eg:

Student (Sid, Sname)
Enroll (Sid, Cid)

```
Select  S.Sid
  From  Student AS S, Enroll AS E
 Where  (S.Sid = E.Sid)
```

Renaming of table

eg: Renaming of attribute

Student (Student_id, Student_name)
Enroll (Student_id, Cause_id)

```
Select   Student.Student_id  AS Sid
From     Student, Enroll AS E
Where    ( Sid = E.Student_id)
```

#e.g. Consider a database that has the relation schema EMP (EmpId, EmpName, and DeptName).

An instance of the schema EMP and a SQL query on it are given below:

EmpId	Emp Name	DeptName
1	XYA	AA
2	XYB	AA
3	XYC	AA
4	XYD	AA
5	XYE	AB
6	XYF	AB
7	XYG	AB
8	XYH	AC
9	XYI	AC
10	XYJ	AC
11	XYK	AD
12	XYL	AD
13	XYM	AE

EC

Count(Emp-id)

DeptName	NUM
AA	4
AB	3
AC	3
AD	2
AE	1

```
SELECT AVG(EC.Num)
FROM EC
WHERE (DeptName, Num) IN
    (SELECT DeptName, COUNT(EmpId) AS EC(DeptName, Num)
                                    FROM EMP
                                    GROUP BY DeptName)
```

The output of executing the SQL query is _____

Subquery (Independent query)

↓

i.e. it will be executed first and an alias with name EC is created

On EC table, outer query will execute

#e.g. Consider a database that has the relation schema EMP (EmpId, EmpName, and DeptName).
An instance of the schema EMP and a SQL query on it are given below:

```
SELECT AVG(EC.Num)
FROM EC
WHERE (DeptName, Num) IN
    (SELECT DeptName, COUNT(EmpId) AS EC(DeptName, Num)
     FROM EMP
     GROUP BY DeptName)
```

The output of executing the SQL query is _____

o/p of this subquery
EC

DeptNam	Num
AA	4
AB	3
AC	3
AD	2
AE	1

$$\frac{4+3+3+2+1}{5} = \frac{13}{5} = 2.6$$

Ans



Topic : WITH clause

WITH clause is used to make the query more understandable

output of a subquery

The WITH Clause is mainly used to provide a subquery block a name that can be referenced in several places within the main SQL query.

eg:

```
WITH new-table_name (list of Attributes of new_table) AS { .... Sub-query ..... }
Main query {
Select ....
}
```

In this main query whenever the output of the subquery is required we can simply refer new_table_name table.

New_table_name is a name assigned to the output of this subquery

#e.g. Consider the following database table named water_scheme

water_scheme		
scheme_no	district_name	capacity
1	Ajmer	20
1	Bikaner	10
2	Bikaner	10
3	Bikaner	20
1	Churu	10
2	Churu	20
1	Dungargarh	10

The number of tuples returned by the following SQL query is _____

```
with total(name, capacity) as
    select district_name, sum(capacity)
    from water_schemes
    group by district_name
with total_avg(capacity) as
    select avg(capacity)
    from total
select name from total, total_avg
    where total.capacity >= total_avg.capacity
```

Subquery-1

Sub-query 2

Main query

total

(name) dist_name	Capacity Sum (Capacity)
Ajmer	20
Bikaner	40
Churu	30
Dungargarh	10

total_avg

Capacity Avg (Capacity)
25

#e.g. Consider the following database table named water_scheme

From total , total_avg
|||
total X total_avg

	total.name	total.Capacity	total_avg.Capacity
X	Ajmer	20	25
✓	Bikaner	40	25
✓	Churu	30	25
X	Dungargarh	10	25

The number of tuples returned by the following SQL query is 2

```
with total(name, capacity) as
        select district_name, sum(capacity)
        from water_schemes
        group by district_name
with total_avg(capacity) as
        select avg(capacity)
        from total
select name from total, total_avg
        where total.capacity >= total_avg.capacity
```

O/P =

name
Bikaner
Churu



Topic : Comparison with NULL

- NULL is a non-zero value.
- No two NULL are equal.

Consider the following Employee table

EMP =

Eid	Ename	Pass_No
E_1	A	123xyz
E_2	B	247ABC
E_3	B	NULL

Query: Retrieve Eid of the Employee with NO passport

```
Select Eid
From Emp
Where Pass-no = 0
```

```
Select Eid
From Emp
Where (Pass-no = NULL)
```

"IS" is the operator to Compare with NULL values

- Complement of "IS" is "IS NOT".

```
Select Eid
From Emp
Where Pass-No IS NULL.
```



2 mins Summary

- **Topic** Execution of Nested queries
- **Topic** AS clause
- **Topic** WITH clause
- **Topic** Comparison with NULL
- **Topic** Comparison with regular expression

Slide

THANK - YOU